

Expt 6	Finite Word length Effects
Date:	6.2 Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation and Rounding Techniques in MATLAB

Code:

```

clc; clear; close all
n=0:79;
x=sin(2*pi*n/40);
q=8;
xt=floor(x*q)/q;
xr=round(x*q)/q;et=x-xt;
er=x-xr;
% Fig 1
figure
plot(n,x,'b',n,xr,'r','LineWidth',1.5)
grid on
title('Original vs Quantized')
legend('Original','Quantized')
% Fig 2
figure
plot(n,x,'b',n,xt,'r',n,xr,'g','LineWidth',1.5)
grid on
title('Truncation vs Rounding')
legend('Original','Truncated','Rounded')
% Fig 3
figure
stem(n,et,'r')
hold on
stem(n,er,'g')
grid on
title('Quantization Error')
legend('Truncation','Rounding')
% Fig 4
figure
bar([mean(et.^2) mean(er.^2)])
set(gca,'XTickLabel',{'Truncation','Rounding'})
grid on
title('MSE Comparison')
ylabel('MSE')

```

Outputs:



